Exact Flux-Band Spectra of Hamiltonian $\mathrm{SU}(N)$ Lattice Gauge Theory

The homological carrier, its all-rank coefficients, and the grading of the strong-coupling series

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Abstract

We study the strong-coupling spectrum of the Kogut–Susskind Hamiltonian for pure $\mathrm{SU}(N)$ gauge theory on a cubic spatial lattice, organized around a homological object fixed before any perturbation theory: the kernel of the plaquette boundary map, $Z_2 = \ker \partial_2$.

Projection to the fundamental charge-odd one-plaquette manifold gives a three-orbital Bloch Hamiltonian whose second-order hopping is not an engineered flat-band model. Non-abelian group integration generates the oriented edge–face boundary Gram operator $BB^\dagger$ with $B = \partial_2^\dagger$, and the amplitude in front of it is the exact rational function

$$t_N = \frac{2N(N^2-4)}{(N^2-1)(2N^2-1)(4N^2-9)} > 0 \qquad (N \geq 3),$$

so that colour dynamics fixes the coefficient and cellular geometry fixes the kernel and the spectrum. The four channel weights producing t_N are order-two Weingarten values, not an isotropy assumption; the vanishing at $N = 2$ is an identity between the parallel and antiparallel channel sums, not a smallness. The projected spectrum carries one momentum-independent level and a doubly degenerate branch shifted by $t_N u^2 q(k)$, and the flat carrier has exact multiplicity $L^3 + 2$ for $L \geq 3$.

At fourth order the effective operator splits into Hodge terms plus one geometric term coupling the carrier to its complement, with support $\{I, U, S, S^2, R\}$. Its off-axis coefficient was disputed between two exact rationals and is decided here by derivation: three independent implementations and the original pipeline's own recovered intermediate ledger agree on every cluster but one, and the exception is diagnosed as an enumeration shortfall of sixteen missing insertion orderings, giving $C_{\text{shp}} = C_{\text{historical}} + 25/1024$ exactly.

Every coefficient is computed as a rational function of the rank rather than interpolated through ranks: we run the exact engine over $\mathbb{Q}(N)$, which removes the degree bound that rational interpolation would otherwise require. This yields two grading theorems. At even orders the one-plaquette C-parity splitting obeys $S_m(N) = c_m N^{-(2m-1)}(1 + O(N^{-2}))$ for $m \leq 10$, with $c_2 = -4$, $c_4 = -32$, $c_6 = -1748/3$, $c_8 = -123332/9$, $c_{10} = -49593808/135$, and the scaling is a cancellation theorem: four intermediate channels each of order $N^{-(m-1)}$ have leading coefficients in the ratio $(-4, +2, +1, +1)$, summing to zero. At odd orders the coefficients vanish identically for even N; for odd N the first survivor is order $N - 2$, a single closed-form vertex.

At sixth order we give the complete eighteen-word folded operator and prove, by exhibiting an explicit point of the Laurent ring where q vanishes and the numerator does not, that the sixth-order band coefficient cannot lie in the fourth-order shape span — whatever the sixth-order dynamics turns out to be.

These are finite-order, finite-volume statements about coefficients of a formal expansion, established exactly. They are not a continuum glueball theorem and not a mass-gap theorem. Every statement carries its machine-verification tier and the command that re-establishes it.

Contents

1 Setting and conventions

1.1 The Hamiltonian

Work on the cubic lattice $\mathbb{Z}^3$ with gauge group $G = \mathrm{SU}(N)$, one group element U_ℓ per oriented link. The Kogut–Susskind Hamiltonian is

$$H \;=\; H_0 + V, \qquad H_0 = \tfrac{1}{2}\sum_{\ell} E_\ell^2, \qquad V = -\,u\,W, \tag{1.1}$$

where E_ℓ^2 is the quadratic Casimir of the left-invariant electric field on link ℓ, and W is the sum of plaquette characters in both orientations,

$$W = \sum_p (\chi_p + \bar{\chi}_p), \qquad \chi_p = \mathrm{Tr}\prod_{\ell\in\partial p} U_\ell. \tag{1.2}$$

Strong-coupling perturbation theory is the expansion in u around the electric vacuum, organized as degenerate perturbation theory — a des Cloizeaux effective operator — on the sector of interest.

1.2 The coupling, and one erratum that matters

The canonical coupling variable throughout is

$$u \;=\; \frac{\beta_N}{2N}, \tag{1.3}$$

whose SU(3) specialization is $u = \beta_3/6$.

A warning, because it has corrupted transcriptions of this material before. An archived line reading $Y = 2\beta/3 = 4u$ is a definition-label erratum, not a change of variable. The coefficients printed alongside it were *already* in the canonical coordinate (1.3). Applying a 4^r rescaling at order r to "correct" them corrupts every order, and the repository carries an explicit check that the printed towers reproduce their canonical values verbatim.

For rank-uniform statements the natural variable is the planar one,

$$\tau \;=\; \frac{2u}{N^2} \;=\; \frac{\beta_N}{N^3}, \tag{1.4}$$

in which the grading theorems of §5 take their simplest form: every even order of the one-plaquette splitting then carries the same single power of N.

1.3 Two projections that must not be conflated

Two distinct projections appear throughout, and several contradictions in this program's own register were naming collisions between them.

The *band* projection is onto the degenerate strong-coupling subspace carrying the excitation of interest — for the flux band, the cellular subspace of §2. Coefficients anchored to it are band-kernel quantities.

The *physical* projection is onto the gauge-invariant, vacuum-subtracted spectrum at a definite momentum. Coefficients anchored to it are physical quantities.

These are related by translation-local scalar shifts, and such a shift changes neither the centered operator, nor its eigenvectors, nor the bandwidth. It does change the scalar. Consequently the band-kernel anchor and the physical Γ-point coefficient are two differently anchored coordinates, not two competing estimates of one quantity, and they are written $q^{(4)}_{\text{band}}$ and $m^{(4)}_{\Gamma}$ respectively. Writing either as "m_4" regenerates a contradiction that does not exist, and the repository's search tooling refuses the name for that reason.

1.4 Exactness

Every coefficient displayed in Part **??** is an exact rational, or an exact rational function of N. Where a quantity is known here only numerically it carries an explicit numerical marker in the source registry and is never displayed as though exact. The single most dangerous defect available in this material is a float that reads as exact, and the boundary between the two is machine-guarded rather than maintained by care.

2 The homological carrier

2.1 Where the flat band comes from

Project the Kogut–Susskind Hamiltonian (1.1) onto the fundamental, charge-odd, one-plaquette manifold. In three spatial dimensions a fundamental plaquette excitation carries one of three face orientations, and at second order neighbouring faces communicate through a shared link. The projected problem is therefore a three-orbital Bloch Hamiltonian, and it looks at first like a tight-binding model.

It is not an engineered one. The signs of the matrix elements are fixed by the orientation of the plaquette boundary and by charge conjugation, and non-abelian group integration generates, exactly, the oriented edge–face boundary Gram operator

$$B(k)B(k)^\dagger, \qquad B = \partial_2^\dagger, \tag{2.1}$$

with the coefficient of §3 in front of it. The structure of the result is a clean separation:

$$\boxed{\text{colour dynamics} \longrightarrow t_N, \qquad \text{cellular geometry} \longrightarrow BB^\dagger.} \tag{2.2}$$

Representation theory fixes the amplitude; the cubical chain complex fixes the kernel and the spectrum. Neither borrows from the other.

2.2 The projected spectrum

Theorem 2.1 (Second-order projected spectrum)**.** *The projected second-order spectrum consists of one momentum-independent level together with a doubly degenerate branch shifted by $t_N u^2 q(k)$, where*

$$q(k) \;=\; 4\sum_{j=1}^{3} \sin^2(k_j/2). \tag{2.3}$$

The flat level is exact and it is exact for a reason: the flat carrier is $\ker \partial_2$.

Theorem 2.2 (The carrier and its multiplicity)**.** *Elementary cube boundaries are compact localized states of the carrier; three wrapping sheets complete it on the torus. For $L \geq 3$ the exact multiplicity is*

$$(L^3 - 1) + 3 \;=\; L^3 + 2. \tag{2.4}$$

Graph id: `RESULT:UNIVERSAL_CELLULAR_HODGE_SPECTRUM`

The two contributions are structurally different and the distinction matters later. The $L^3 - 1$ cube boundaries are local and finitely supported. The three sheets are not: they wrap the torus, and they are the entire zero-momentum content of the carrier. §?? returns to the consequence — that no finitely supported exactly-flat local operator can see the Γ point at all.

2.3 Third order, and the coefficients that survive

For SU(3), exact third-order perturbation theory stays inside the same boundary-factorized algebra. It does not generate a new operator structure; it renormalizes the coefficient in front of the same $BB^\dagger$:

$$H_{\text{eff},-} \;=\; E_{\text{flat}}(u)\, I \;+\; \left(\frac{5}{612}u^2 + \frac{1975}{124848}u^3\right) BB^\dagger \;+\; O(u^4), \tag{2.5}$$

with

$$E_{\text{flat}}(u) \;=\; \frac{8}{3} + u + \frac{11}{306}u^2 - \frac{109151}{249696}u^3. \tag{2.6}$$

That (2.5) remains boundary-factorized through third order is the non-trivial part. It is what makes the fourth order the first place where the carrier can couple to its complement, and §4 is about exactly that coupling.

2.4 What is and is not novel here

This program does not claim that incidence flatness, cellular compact localized states, or homological counting of zero modes are new. All three have a substantial prior literature — line-graph constructions, compact localized states, the necessity of non-contractible states on periodic systems, rectangular factorizations and index arguments, and closely related closed-surface structures in two-form spin liquids — and a recent independent construction treats face-graph and higher-cell flat bands directly.

What is claimed is that the flatness here is *dynamical*: it is produced by exact strong-coupling matrix elements of non-abelian SU(N) Hamiltonian lattice gauge theory, with the amplitude fixed by group representation theory and the kernel fixed by the cubical chain complex, as in (2.2). The flat band is not put in; it comes out.

Scope 2.3. Theorem 2.1 and (2.5) are finite-volume, finite-order statements about a projected flux sector. They are not a continuum glueball theorem and not a mass-gap theorem, and the source manuscript says so in its own abstract.

3 Second order at every rank

This section is the algebraic base of the whole program. Everything in it is proof-checked, and its conclusion — that the band exists for $N \geq 3$ and not for $N = 2$, with an exact amplitude — follows from representation theory with no input from any expansion.

3.1 The shared-link weights are Weingarten values

Two plaquettes contributing at second order either share a link or do not. The amplitude for the shared-link case was for a long time taken in this program with an isotropy premise: that the four contributing representation channels carry weights in a fixed ratio. That premise is not needed. It is a consequence of the order-two Weingarten values.

Proposition 3.1 (The shared-link weights, from representation theory alone)**.** *Let two plaquettes share exactly one link. Then:*

1. *The six non-shared links collapse. Each plaquette contributes a product of three independent Haar links, and a product of independent Haar matrices is Haar. So the two-plaquette amplitude reduces to* $\mathrm{Tr}(AU)\ \mathrm{Tr}(BU^{\pm 1})$ *with A and B independent Haar, and integrating them out leaves a pure degree-*$(2,2)$ *moment of the shared link U.*

2. *Two such moments settle both families. With $M_{\text{direct}} = N^2$ and $M_{\text{cross}} = N$, the like family splits as $(N+1)/(2N)$ and $(N-1)/(2N)$; and the singlet component of $U_{ij}\overline{U_{lk}}$ is $\delta_{il}\delta_{jk}/N$, of squared norm 1, giving $1/N^2$.*

All four weights are exactly d_R/N^2, where d_R is the dimension of the channel.

Graph id: `theshared-linkweightsareWeingarten,notanisotropyassumption` Reproduce: `workhouse verify -only 'the shared-link weights are Weingarten, not an isotropy assumption'`

[T1 exact]. The relevant Weingarten pair is the inverse of the S_2 Gram matrix and the index sums are explicit, so the check imports nothing from the corpus. Registered as the discharge of the gap that had carried this premise as the one unproved input of the second-order chain; it was opened and closed on the same day, 28 August 2026, once the question was asked in the right form.

3.2 The four channels in closed form

Write $C_F = (N^2-1)/(2N)$ for the fundamental Casimir. Each channel R contributes a resolvent weight $-(d_R/N^2)/(C_F + \frac{1}{2}\Delta C_2(R))$, and all four evaluate in closed form.

Theorem 3.2 (Per-channel resolvent weights)**.**

$$w_{\mathbf{1}} = -\frac{1/N^2}{C_F} = -\frac{2}{N(N^2-1)}, \tag{3.1}$$

$$w_{\mathrm{Adj}} = -\frac{(N^2-1)/N^2}{C_F + N/2} = -\frac{2(N^2-1)}{N(2N^2-1)}, \tag{3.2}$$

$$w_{\mathrm{Sym}} = -\frac{N(N+1)/2N^2}{C_F + \frac{(N-1)(N+2)}{2N}} = -\frac{N+1}{(N-1)(2N+3)}, \tag{3.3}$$

$$w_{\Lambda} = -\frac{N(N-1)/2N^2}{C_F + \frac{(N+1)(N-2)}{2N}} = -\frac{N-1}{(N+1)(2N-3)}. \tag{3.4}$$

Graph id: `LEAN:resolventWeight_singlet,LEAN:resolventWeight_adjoint,LEAN:resolventWeight_sym,LEAN:resolventWeight_antisym`

[T0 Lean] — all four, at `lean/Workhouse/Basic.lean:72,80,89,104`. Reproduce with `make lean`.

3.3 The two family sums

The channels group into a *mixed* family $\{\mathbf{1}, \mathrm{Adj}\}$ and a *like* family $\{\Lambda^2 F, \mathrm{Sym}^2 F\}$, and these are the antiparallel and parallel channel sums.

Theorem 3.3 (Family sums)**.**

$$A_N \;=\; w_{\mathbf{1}} + w_{\mathrm{Adj}} \;=\; -\frac{2N^3}{(N^2-1)(2N^2-1)}, \qquad B_N \;=\; w_{\mathrm{Sym}} + w_{\Lambda} \;=\; -\frac{4N(N^2-2)}{(N^2-1)(4N^2-9)}. \tag{3.5}$$

Graph id: `LEAN:mixed_family_sum,LEAN:like_family_sum`

[T0 Lean], at `lean/Workhouse/Basic.lean:118,129`. The mixed sum is obtained from the dimension and Casimir table rather than by transcription from any source document, which is the point: the family sums are re-derived, not quoted.

3.4 The hopping amplitude

Theorem 3.4 (The all-rank second-order hopping amplitude)**.**

$$t_N \;=\; B_N - A_N \;=\; \frac{2N\,(N^2-4)}{(N^2-1)(2N^2-1)(4N^2-9)}. \tag{3.6}$$

It is strictly positive for every $N \geq 3$, and

$$t_2 = 0, \qquad t_3 = \tfrac{5}{612}. \tag{3.7}$$

Graph id: `CONST:t_N` Reproduce: `workhouse why t_N`

[T0 Lean] for the rank law and for both special values (`lean/Workhouse/Basic.lean`: `rank_law_numerator`, `hopping_two`, `hopping_three`). Clearing denominators, the channel difference has numerator exactly $2N(N^2-4)$.

Two consequences deserve to be stated separately, because both are structural rather than numerical.

Corollary 3.5 (SU(2) has no band at this order)**.** *The factor N^2-4 in (3.6) vanishes at $N=2$. The exclusion of* SU(2) *is therefore not a smallness but an identity: the parallel and antiparallel channel sums coincide exactly at rank two.*

Graph id: `LEAN:hopping_two`

Corollary 3.6 (The planar approach is from below)**.** *With $D = (N^2-1)(2N^2-1)(4N^2-9)$,*

$$D - 4N^3 \cdot 2N(N^2-4) \; = \; 2N^4 + 31N^2 - 9 \; > \; 0 \qquad (N \geq 1), \tag{3.8}$$

so $4N^3 t_N = 1 - (2N^4+31N^2-9)/D < 1$. Hence $t_N < 1/(4N^3)$ for every N, and $t_N \sim 1/(4N^3)$: the N^{-3} cancellation is exact and what it leaves is a strictly positive deficit.

Graph id: `LEAN:hopping_deficit_numerator`

[T0 Lean], at `lean/Workhouse/Basic.lean:48`. Corollary 3.6 is the first appearance of the rank grading that §5 develops: the leading power of N is fixed by a cancellation, and the surviving term is a polynomial one can write down.

3.5 Independent corroboration of the channel weights

The weights of Proposition 3.1 have an external provenance, from work written without knowledge of this program.

An efficient-truncation study of SU(N_c) lattice gauge theory for quantum simulation [21] gives, in its Appendix D, the exact finite-rank plaquette matrix element

$$|M_\rho| = \sqrt{\frac{\dim \rho}{\dim A \, \dim R}}. \tag{3.9}$$

Taking both factors fundamental gives $|M_\rho|^2 = d_\rho/N^2$, which is *identically* the numerator of the weight formula used here. The four shared-link channel weights, and with them A_N, B_N and $t_N = B_N - A_N$, therefore rest on a matrix element an independent group wrote down for an unrelated purpose. That truncation is also the first published one retaining all four shared-link irreducible representations.

The agreement is evidence rather than bookkeeping precisely because neither side knew of the other. It does not make t_N published: the rank law (3.6), the vanishing at $N=2$, and the deficit of Corollary 3.6 are assembled here.

A caution attaches in the other direction. The leading-large-N_c qutrit plaquette Hamiltonian [20] explicitly decouples the charge-odd combination, so the sector in which t_N lives is inert in that model — effectively $t_3 = 0$ at leading order in large N_c, against the finite-N channel-complete value 5/612. That paper is recorded in this program's index as not yet obtained, and the statement here is taken from imported bridge documents rather than checked against the paper itself.

3.6 What this section does not establish

Does not assert. Convergence of the strong-coupling series; anything about the spectrum at finite u beyond second order; and nothing about SU(2) at higher orders, where Corollary 3.5 does not apply. Theorem 3.4 is a statement about one coefficient of a formal expansion, established exactly.

4 Fourth order, and how a disputed coefficient was decided

Fourth order is where the carrier first couples to its complement, and it is where this program spent the largest single block of its effort. The mathematics is one theorem and one number. The number was disputed for months between two irreconcilable values, and the way it was settled is, in this author's view, the more instructive half of the story.

4.1 The structure

The fourth-order effective operator splits into Hodge terms and one geometric term coupling the carrier to its complement. Restricted to range-one operator structures, the shape content collapses: each such structure contributes a fixed shape coefficient, and every one of them gives $B = D = 0$ identically. The tier collapse is simply what range-one looks like.

Theorem 4.1 (Fourth-order support, at SU(3))**.** *The recorded fourth-order kernel has support $\{I, U, S, S^2, R\}$, and the shape data closes in that span. Solved over the whole Brillouin zone in exact rationals rather than fitted, the shape table is determined by three signed amplitudes:*

$$C_{\text{shp}} \;=\; -\tfrac{5}{96} - u - \tfrac{1}{2}(\rho + \pi), \qquad u_2 = 2u, \qquad \nu = -\big(\tfrac{5}{48} + 4u\big), \tag{4.1}$$

where u is the two-hop chain cumulant.

Graph id: `RESULT:TIER_COLLAPSE_ACTUAL_H4_SUPPORT,RESULT:TIER_COLLAPSE_IS_R_DEGREE`

Scope 4.2. "The recorded kernel" means the 189-record historical kernel, which is $N = 3$. The operator splitting is an SU(3) statement; it is not established at general rank, and framing it as an SU(N) identity would be an overreach. The rank-uniform statements of this program are the coefficient laws of §3 and §5, not this splitting.

Clearing the $1/q$ makes the shape ansatz a linear system over Laurent polynomials in e^{ik_j}, which is why the whole zone can be solved exactly instead of sampled.

There is a second, equivalent presentation. In the Hodge form the two-hop weight is absorbed, $\tilde{\pi} = \pi + 2u$, and

$$C_{\text{shp}} \;=\; -\tfrac{5}{96} - \tfrac{1}{2}\big(\rho + \tilde{\pi}\big), \tag{4.2}$$

so that u drops out of C entirely and exactly two shared-link amplitudes remain. Equations (4.1) and (4.2) are the same statement. They must not be combined — writing u and $\tilde{\pi}$ in the same expression double-counts the two-hop weight, and that error has been made.

4.2 The dispute

Two independently recorded kernels — a historical pipeline from June 2026 and a later cold dump — gave different values for the off-axis shape coefficient C_{shp}. Both were exact rationals. Neither could be promoted, because the repository's governing rule is that a dispute closes by derivation or not at all: code may not pick a side, average them, or prefer whichever rational looks more authoritative.

Comparing the two kernels record by record localized the disagreement rather than resolving it. Both have the same six weight orbits and the same normalized shape table; both satisfy the same two relations above. What differs is the skeleton scale and the *signs* of two amplitudes. One scale factor explains 144 of the 189 records to 2×10^{-12}: the rotation bulk is structurally shared, and the dispute is three amplitudes, not two kernels.

4.3 How it was settled

By building a third implementation that shares no primitive with either, and by recovering the historical pipeline's own intermediate ledger.

The third implementation is a Wilson-loop calculus written from scratch: the free Hamiltonian by the Fierz identity as a rewiring of index ports — a like pair swaps wires, an unlike pair is cut and crossed, which is unitarity; Haar integrals by the $U(N)$ Weingarten function as the Gram pseudoinverse on S_n, with $\det U = 1$ inserted on virtual slots for the determinant families; and the resolvent by per-link irreducible projectors, so that every component is an exact eigenvector of H_0 with a known energy. It reproduces the four second-order constants, the two-hop weight with the correct sign on three chain types, and the single-contact and shared-link dressings, in seconds — and then gives the corner cumulant to the digit.

Separately, a survey of the corpus located the historical pipeline's own Stage-3I ledger: 4,221 ordered words whose hash matches the input the kernel copies quote. Its rooted assembly, ported verbatim, reproduces the pinned 189 records exactly. Grouping each word's rooted images by the set of plaquettes it touches then yields the historical kernel's own connected cumulants — a reading the original pipeline never performed on its own data.

Theorem 4.3 (Adjudication of the off-axis coefficient)**.** *The historical kernel and the independent assembly agree on* every *cluster of the rotation record — pair, fourteen chains, two fans, two corners — with identical rationals in both C-parity sectors, except one. The exception is the adjacent-face cube completion. The historical ledger holds* 8 *of the* 24 *insertion orderings, each with exactly the weight the multi-loop model assigns it, namely* $(2, 1, 4, 2, 8, 8, 2, 4)$ *in units of* $1/1536$*, totalling* $31/1536$*; the sixteen it lacks sum to* $-25/512$*. The opposite-face completion in the same ledger has all* 24*.*

Consequently, in the kernel's basis,

$$C_{\text{shp}} \;=\; C_{\text{historical}} + \frac{25}{1024} \;=\; -\frac{13035490122347}{550663802582400} \;=\; -0.0236723\ldots \tag{4.3}$$

Substituted into the 24 *records and solved over the whole zone, this gives* $A = 5/48$ *and* $B = D = 0$ *with no residual.*

Graph id: `C2,ADR:0024` Reproduce: `workhouse why C2`

[T1 exact], suite "the third implementation and the historical ledger": nine exact checks, three of them registered findings.

The shortfall has a closed form at every rank. The adjacent-face completion is

$$c_4^{\text{adj}}(N) = \frac{-106}{N(N^2-1)^3}, \qquad c_4^{\text{adj}}(3) = -\frac{53}{768}, \tag{4.4}$$

in ratio 53/80 to the opposite-face completion $-160/(N(N^2-1)^3)$; the -106 is the primitive -88 plus ten multi-loop orderings at -18.

4.4 Three things this episode establishes

The disagreement was an enumeration shortfall, not a disagreement about mathematics. A pipeline dropped sixteen of twenty-four orderings on one cluster and nowhere else. That is a diagnosable defect, and it was diagnosed ordering by ordering rather than adjudicated by preference.

A basis error can survive validation. An earlier assembly of the same amplitude was correct and was read in the conjugate basis, giving a third value. Every C-odd component of the historical ledger is the negative of the assembly's value in one traversal while every C-even one is identical — the signature of conjugating a single face. The assembly was right; its basis was not. The decision record amends it rather than withdrawing it.

The right answer arrived by a forbidden route first. The value (4.3) is the one an all-rank continuation formula produces at $N = 3$ — and that derivation was retracted, because the substitution it used is not permitted. The retraction stands. That the forbidden route happened to reach the number the legitimate assembly later confirmed is not a vindication of the route, and the register keeps both facts.

Scope 4.4. Both recorded kernels remain in the registry with their verdicts: the historical superseded, the cold dump falsified, beside the proven assembled value. Nothing was deleted. One follow-up remains untried — the pair cluster from the third engine, pure-six family included — which can sharpen this result but not overturn it.

5 Rank grading: cancellation and determinant families

The strong-coupling coefficients of §3 and §4 are exact rational functions of N. That makes a question available which is invisible at any single rank: how does each order of the series scale with the rank, and why?

The answer has two halves. At even orders the scaling is fixed by an exact cancellation among four representation channels. At odd orders the coefficients vanish outright for even N, and for odd N they first appear at order $N-2$ with a closed-form vertex. Neither statement can be seen from numerics at $N = 3$.

5.1 The one-face grading law

Let $S_m(N)$ be the C-parity splitting of the one-plaquette strong-coupling diagonal at order m,

$$S_m(N) \;=\; (\mathrm{odd})_m - (\mathrm{even})_m \;=\; -2\,K_m[0][1]. \tag{5.1}$$

Theorem 5.1 (Single-face planar grading through tenth order)**.** *At even orders $m = 2, 4, 6, 8, 10$ the exact rational function of N satisfies*

$$S_m(N) \;=\; c_m\, N^{-(2m-1)}\big(1 + O(N^{-2})\big), \tag{5.2}$$

with

$$c_2 = -4, \quad c_4 = -32, \quad c_6 = -\tfrac{1748}{3}, \quad c_8 = -\tfrac{123332}{9}, \quad c_{10} = -\tfrac{49593808}{135}. \tag{5.3}$$

Writing $u = N^2\tau/2$ as in (1.4), the leading *part of the order-m contribution is* $(c_m/2^m)\,N\tau^m$. *Every even order therefore carries the same single power of N, with*

$$b_2 = -1, \quad b_4 = -2, \quad b_6 = -\tfrac{437}{48}, \quad b_8 = -\tfrac{30833}{576}, \quad b_{10} = -\tfrac{3099613}{8640}, \qquad b_m = c_m/2^m. \tag{5.4}$$

Graph id: `RESULT:SINGLE_FACE_PLANAR_GRADING`

Recorded `proven` with `output-certified` evidence. The exponent and the coefficients are exact limits of exact rational functions of N, not numerical extrapolations. An independent integer-rank engine reproduces c_2, c_4, c_6, c_8 by extrapolation to within 1.4×10^{-5} relative — a cross-check by a different method, not the source of the values.

The word *leading* in Theorem 5.1 is doing work and should not be dropped. Each S_m is an exact rational function of N with relative corrections in N^{-2}, so the reduction to τ is a planar-limit statement, not an identity at finite rank. Explicitly, at second order

$$S_2(N) = -4N^{-3} - 4N^{-7}, \qquad \text{so} \qquad \frac{S_2\,u^2}{N} = -\tau^2 - \frac{\tau^2}{N^4}. \tag{5.5}$$

The single-face band per unit N is therefore a function of τ alone *as* $N \to \infty$ at fixed τ, and not at finite N. A statement of the second form would be false, and an earlier internal summary of this result made it.

5.2 The grading is a cancellation theorem

Equation (5.2) is a much smaller number than its constituents, and the reason is exact.

Theorem 5.2 (Four-channel cancellation)**.** *Decomposing S_m by the intermediate state of the perturbed vector gives exactly four contributing channels at every even m from 4 to 10: the singlet $((),())$, the adjoint $((1),(1))$, and the two-box states $((2),())$ and $((1,1),())$. Each is of order $N^{-(m-1)}$, with leading coefficients*

$$(-4,\ +2,\ +1,\ +1) \times 2^{\,m/2-2}, \tag{5.6}$$

which sum to zero*. At $m = 2$ only the singlet and the adjoint contribute, at -2 and $+2$, and these also cancel.*

Graph id: `RESULT:SINGLE_FACE_PLANAR_GRADING`

So the individual channels are of order $N^{-(m-1)}$ and their leading behaviour annihilates, leaving (5.2) at $N^{-(2m-1)}$ — two further orders down in the N^{-2} expansion. At $m = 4$ this is the statement that channels of order N^{-3} conspire to produce a band of order N^{-7}.

This is why the grading is stated as a theorem about a mechanism rather than as an observed power. A power law read off from fitted data at several ranks would carry no information about *why* the leading terms are absent, and would give no reason to expect (5.2) to persist. The cancellation does.

Scope 5.3. Theorem 5.1 concerns the *one-plaquette* diagonal splitting at even orders $m \le 10$. It is not the cluster cumulant of the multi-face conjecture, which is not discharged by it: the one-face case at sixth order was already established separately, and the shared-link pair route carries the multi-face statement. The general-m law remains a conjecture. All statements are about coefficients at fixed order, not about convergence.

5.3 Odd orders are determinant families

Theorem 5.4 (Centre parity)**.** *For* even N*, every odd-order matrix element of the des Cloizeaux effective operator of the strong-coupling band vanishes identically — on every cluster, in both C-parity sectors, at every order.*

For odd N*, an odd order m vanishes unless $m \geq N-2$. At $m = N-2$ the only surviving element is the one-plaquette vertex*

$$H_{N-2}(\bar{F}, F) \;=\; -\,\frac{\big(N/(N+1)\big)^{N-3}}{\big((N-3)!\big)^{2}}, \tag{5.7}$$

which is -1 at $N = 3$ and $-\frac{25}{144}$ at $N = 5$. Hops and leakages vanish at that order.

Graph id: `RESULT:ODD_ORDER_CENTRE_PARITY`

The mechanism is the centre. The SU(N) Haar integral of a word vanishes unless every link's net flux is 0 mod N; on a single plaquette the four links are private, so the intermediate states are characters with energy $2C_2(R)$, and the flux condition at odd order can be met only once enough boxes have accumulated to close modulo N. That threshold is $N-2$, and it is why the first odd order is a function of the rank rather than a fixed number.

Scope 5.5. Statements about coefficients of the strong-coupling series at fixed order, not about convergence. On a periodic lattice the link-set lemma requires even extents. The even orders and the overlap theorem are not addressed here.

5.4 Why this refuted a claim

The register records a claim that centre-only circuits are dynamically dark — that configurations carrying only centre flux cannot contribute to the dynamics. Theorem 5.4 and its fifth-order companion refute it: the exact fifth-order determinant correction $\delta c_{5,\mathrm{det}}$ is nonzero. The claim was falsified by an exact computation rather than by an estimate, which is the only way a claim of that shape can be settled.

6 The symbolic-rank engine

Every closed form in §3–§5 is a rational function of N computed as such, not interpolated through ranks. This section explains how, because the method is the most transferable thing in the program and because getting it wrong is the default.

6.1 The problem with sweeping ranks

The natural procedure is to run an exact engine at $N = 3, 4, 5, \ldots$ and fit a rational function. That is what was done first. An assembled cluster cumulant β_N agreed with a candidate closed form $P_{17}(N^2)/(N\,R_{20}(N^2))$ at every rank from 4 to 70: sixty-seven exact agreements between exact rationals.

Sixty-seven exact agreements do not prove the identity. Rational interpolation is unique only once the degrees are bounded, and without a degree bound the agreements constrain nothing about $N = 71$. The review of that work made exactly this objection — no "for all N" without a degree bound — and it was correct.

A degree bound could have been argued: Weingarten denominators, energy denominators, and a count of vertices together bound the degree, and the sixty-seven agreements would then prove the identity by uniqueness. That argument was not made. What was done instead makes it unnecessary, and is better.

6.2 Running the engine over $\mathbb{Q}(N)$

Inspect where the rank actually enters the exact engine. It enters in exactly two places that are not rational arithmetic:

- the charge predicate, charge $\equiv 0 \bmod N$;
- the flux-difference predicate, $|\text{diff}| \leq N$.

Everywhere else the rank appears only through rational expressions in $C_F = (N^2-1)/(2N)$, in $1/(2N)$, and in N^{closed}.

So make the two predicates overridable and promote everything else to the rational function field:

- scalars become elements of $\mathbb{Q}(N)$, carried as a numerator and a monic denominator in exact polynomial arithmetic;
- the Weingarten function becomes the symbolic inverse of the Gram matrix N^{cycles} on S_n;
- the single-link spectra become the $\mathrm{SU}(N)$ quadratic Casimirs of the irreducible representations that link's flux content can carry.

A cumulant then emerges as *one rational function of N*, computed rather than fitted. The per-rank engine is otherwise untouched and its own test suite still holds — which is the check that the symbolic run is the same calculation and not a different one.

Theorem 6.1 (Closed forms over $\mathbb{Q}(N)$)**.** *The fourth-order cluster cumulants, the two-hop weight in both C-parity sectors, and the assembled β_N are exact elements of $\mathbb{Q}(N)$ produced by the symbolic engine. The identity $\beta_N = P_{17}(N^2)/(N\,R_{20}(N^2))$ holds as an identity of rational functions.*

Graph id: `ADR:0029`

The final polynomial identity is promoted to [T0 Lean], and it is worth saying exactly which identity that is. The corpus *states* $\beta_N = P_{17}(N^2)/(NR_{20}(N^2))$ and never derives it. What Lean checks is that the closed forms the symbolic engine returns for the three cumulants and the adjacent-face cube completion *sum to* the corpus's coefficient — a polynomial identity between the engine's output and the corpus's formula. It is not a Lean derivation of the cumulants, and the Lean source says so in its own docstring. This document repeats the qualification because an internal summary of the same fact dropped it.

Every eigencomponent of every resolvent is verified to be an exact eigenvector of every single-link H_0, symbolically — a check that has no per-rank analogue, because at a fixed rank a coincidence of numbers is indistinguishable from an identity.

6.3 Why this is the right shape of argument

The methodological content generalizes beyond this program.

An identity between rational functions can be established in two ways: verify it at enough points and bound the degrees, or compute in the function field. The first requires an argument about the size of the objects, which is a second piece of mathematics and is where the difficulty hides. The second requires only that the algorithm's dependence on the parameter be rational — which is a property one can *inspect*, predicate by predicate, rather than estimate.

The title of the decision record is the summary: the degree bound is proved by removing it.

Scope 6.2. The symbolic engine is valid at ranks where the generic partition labelling applies. Exceptional low ranks, where representations degenerate, are handled separately and are the reason SU(2) and in places SU(4) carry their own entries in the register.

This degree bound must not be confused with the *other* degree bound in this program's history — the proposed kinematic mechanism for the fourth-order tier collapse, which was retracted (§4, §**??**). That one was withdrawn because it was wrong. This one was removed because it became unnecessary.

7 Sixth order: a shape that cannot be removed

The fourth-order shape data closes in a five-element span. The question this section answers is whether the sixth order stays inside it. It does not, and the proof is a divisibility argument that needs no knowledge of the sixth-order dynamics at all.

7.1 Conventions

In the Bloch variables z_j write

$$a_j = 2 - z_j - z_j^{-1}, \qquad q = \sum_j a_j, \qquad e_2 = \sum_{i<j} a_i a_j, \qquad e_3 = a_1 a_2 a_3, \tag{7.1}$$

and let $U = \psi\psi^*$ with $\psi^*\psi = q$, $P_c = U/q$, $Q_c = I - P_c$, $S = L_\downarrow - 4I$, and R the cross-plane part of S.

Two warnings, both of which have caused errors here. The invariants e_2, e_3 are Bloch invariants, *not* representation Casimirs; and U is the up-Laplacian, not its normalized projector, so its kernel is the complementary plane rather than the carrier line. Cleared identities hold at $q = 0$; normalized carrier formulas require $q > 0$.

7.2 The complete folded operator

Let P_E be the isolated electric model-space projector, $D = Q_E/(E_0 - H_0)$, and V_0 the perturbation. If $P_E V_0 P_E = aP_E$, set $V = V_0 - aI$ *on the whole Hilbert space*: shifting the complementary block as well is essential. With $\chi_0 = P_E$, intermediate normalization gives the exact recursions

$$K_n = P_E V \chi_{n-1}, \qquad \chi_n = DV\chi_{n-1} - D\sum_{j=1}^{n-1} \chi_{n-j} K_j, \tag{7.2}$$

and with $M = P_E + \chi^*\chi$ the canonical Hermitian effective operator is $M^{1/2} K M^{-1/2}$. Writing

$$A_m = P_E V (DV)^{m-1} P_E, \qquad C_2 = P_E V D^3 V P_E, \tag{7.3}$$

and B_m for the sum of A_m with exactly one of its $m - 1$ slots replaced by D^2:

Theorem 7.1 (The eighteen-word sixth-order fold)**.**

$$H_6 = A_6 - \tfrac{1}{2}\{A_4, B_2\} - \tfrac{1}{2}\{A_3, B_3\} - \tfrac{1}{2}\{A_2, B_4\} + \tfrac{1}{2}\{A_2^2, C_2\} + \tfrac{3}{8}\{A_2, B_2^2\} + \tfrac{1}{4}B_2A_2B_2. \tag{7.4}$$

Expanded, this is exactly eighteen words in P_E, V, D. The same recursion reproduces the established $H_4 = A_4 - \tfrac{1}{2}\{A_2, B_2\}$.

Graph id: `RESULT:G9_HERMITIAN_FOLDED_SUPPORT`

The $\{A_3, B_3\}$ term is present and it matters: dropping odd electric histories gives an incomplete formula. No odd order is dropped anywhere in the recursion. Independent rational matrix checks verify both the invariant-subspace equation and the normalization through degree six, with a rank-two P_E and non-commuting effective coefficients.

Does not assert. Equation (7.4) is the universal folded operator support in this convention. It does not evaluate Wilson plaquette matrix elements, establish a specific support, or carry out connected multi-plaquette rooted subtraction. Those require the actual electric and Haar implementation on each support.

7.3 Exact word symbols

In a polynomial carrier gauge, exact symmetrization gives each word its cleared symbol in $\mathbb{Q}[q, e_2, e_3]$. The table below is computed, not inferred, and all forty words of length zero through three agree coefficient by coefficient with independently implemented spatial Laurent operators.

Word	$\sigma(W)$	Word	$\sigma(W)$
I	q	RR	$qe_2 + 3e_3$
U	q^2	RUR	$4e_2^2$
S	$-4q$	RSR	$(q-4)(qe_2 + 3e_3) - 4e_2^2$
S^2	$16q$	RRR	$-2e_2^2 - 2qe_3$
R	$-2e_2$	URR, RRU	$q^2e_2 + 3qe_3$
UR, RU	$-2qe_2$		

A word census must retain coefficients and combine them before testing shape membership: $\sigma(RUR + 2\,RRR) = -4qe_3$, although both individual words contain e_2^2. Support is not unique modulo operator identities, and the invariant object being tested is the complete carrier polynomial. Missing direct input is represented as unknown, not as zero.

7.4 The band coefficient at sixth order

At fixed nonzero momentum, with scalar anchors removed,

$$H_{\text{eff}}(u) = u^2 t_3\, q\, Q_c + u^3 H_3 + u^4 H_4 + u^5 H_5 + u^6 H_6 + O(u^7). \tag{7.5}$$

The recorded SU(3) third-order factorization gives $Q_c H_3 P_c = 0$, so the off-diagonal block first appears at order u^2 in the scaled operator, from H_4. Expanding the Schur complement for the scalar carrier branch,

$$\lambda_6 = \frac{\sigma(H_6)}{q} - \frac{q\,\sigma(H_4^2) - \sigma(H_4)^2}{t_3\, q^3}. \tag{7.6}$$

This is not an assumption that H_5 vanishes. H_5 first couples to H_4 at order u^7 in H_{eff}, as does the uH_3 correction to the complementary inverse, and unknown off-diagonal H_6 contributes later still. The decoupling of H_3 is an explicit recorded input rather than an assumption, and any scalar vacuum subtraction is part of H_6 and changes nothing.

Evaluating all twenty-five ordered pairs from the actual H_4 support $\{I, U, S, S^2, R\}$, including their projected subtraction, gives zero for twenty-four of them. Only RR survives, in agreement with direct composition of the full assembled spatial kernel. With

$$\kappa = \frac{4C_{\text{shp}}^2}{t_3} = \frac{169924002729806205028788409}{619343593697933825385600000} > 0, \qquad t_3 = \tfrac{5}{612}, \tag{7.7}$$

the known dynamical mixing support is $-(\kappa/q)\,RR + (\kappa/q^2)\,RUR$, with normalized expectation

$$\lambda_6^{\text{mix}} = -\kappa\left(\frac{e_2}{q} + \frac{3e_3}{q^2} - \frac{4e_2^2}{q^3}\right). \tag{7.8}$$

Does not assert. The RUR here arises from inserting $P_c = U/q$ in the projected subtraction. It is not evidence that a direct six-insertion electric history carries the local word RUR, and the ratio $1 : 3 : -4$ belongs to this mixing term rather than to the full coefficient.

7.5 Non-cancellation

Theorem 7.2 (An extra rational shape survives every local direct term)**.** *Assume the order-six electric-shell effective operator is a finite-range, translation-covariant stencil with Laurent-polynomial entries in this basis, and write $F = \sigma(H_6)$. Then (7.6) gives*

$$q^3\lambda_6 = q^2F - \kappa(q^2e_2 + 3qe_3 - 4e_2^2), \tag{7.9}$$

and q does not divide the right-hand side, for any *Laurent polynomial F.*

Proof. Modulo q, the right-hand side of (7.9) is $4\kappa e_2^2$. Evaluate at

$$z_1 = z_2 = -1, \qquad z_3 = 5 + 2\sqrt{6}, \tag{7.10}$$

all nonzero, so the point lies in the Laurent ring. Since $z_3^{-1} = 5 - 2\sqrt{6}$, we get $a = (4, 4, -8)$, hence $q = 0$ and $e_2 = -48$. The value there is $4\kappa \cdot 48^2 = 9216\,\kappa \neq 0$, and a Laurent polynomial divisible by q would vanish at that point. □

Graph id: `RESULT:G9_LOCAL_SIXTH_NONCANCELLATION`

[T1 exact]. Consequently λ_6 cannot lie in the fourth-order shape span $\{1, q, e_2, 4e_2/q, e_3/q\}$, and no local direct RUR term removes the extra rational shape: a direct $h\,RUR$ contributes $4he_2^2/q$, which cannot cancel an e_2^2/q^3 component. For symmetric F the remainder modulo q^2 is exactly $4\kappa e_2^2 - 3\kappa q e_3$, so both e_3/q^2 and e_2^2/q^3 are present. The e_2/q term alone *can* be cancelled, by $F = \kappa e_2$ — for instance a local $H_6 = -(\kappa/2)R$ — so the $1 : 3 : -4$ ratio is not a constraint on all sixth-order shapes.

Note what the proof does not use: any knowledge of F. The extra shape is forced by the fourth-order data alone, through the Schur complement, whatever the sixth-order dynamics turns out to be.

7.6 A withdrawn derivation

An earlier version of this work enumerated closed rooted walks — 90 four-step unit-edge walks, of which 66 retrace immediately and 24 are planar squares, together with 192 non-retracing six-step walks using all three axes — and read a sixth-order amplitude off that census.

The derivation is withdrawn. Edge walks are not ordered plaquette insertions with electric states, Haar weights, projectors and folds, and no map between them was supplied; the census therefore determines no net coefficient of any of the candidate words. The old amplitude formula was assigned rather than derived. The old carrier function used $\sigma(R) = 0$ and $\sigma(U) = q$, where the actual unnormalized symbols are $-2e_2$ and q^2; it returned a nonzero expression for a defect it separately reported as zero. The implementation now computes the symbols and tests the defect, and the old interface raises an explicit missing-dynamics error rather than returning a number.

What is withdrawn is a claimed derivation, not the possibility of a physical amplitude of that shape. Theorem 7.2 is independent of the census entirely. Computing the direct term on a connected multi-plane support is Problem **??**, and it is the cheapest decisive computation currently open in this program.

8 A rank-uniform local shell construction

This section is a separate result in its own regime, included because it is exact, rank-uniform, and independent of everything above — not because it continues it. The object is the formal local Wilson oscillator expansion around a single plaquette, and the point is that its spectrum admits a finite exact recursion at every perturbative order with coefficients in $\mathbb{Q}[N, N^{-1}]$, needing no Gram-matrix inversion.

Conflating this construction with the flux band of §2–§7, or with the Wilson transfer operator of §**??**, would be an error. They are four different objects — the physical SU(N) transfer band, the SU(3) internal Hamiltonian carrier, the conditional block, and this local oscillator — and no cross-regime continuation between them is asserted anywhere in this document.

8.1 The exact algebra

Let X be a traceless Hermitian $N \times N$ matrix with density proportional to $e^{-\operatorname{Tr}(X^2)}$, write $P_k = \operatorname{Tr}(X^k)$ with $P_0 = N$ and $P_1 = 0$, and use a traceless Hermitian basis normalized by $\operatorname{Tr}(T_a T_b) = \delta_{ab}$. The covariance is

$$\mathbb{E}[X_{ab} X_{cd}] = \tfrac{1}{2}\left(\delta_{ad}\delta_{bc} - \tfrac{1}{N}\delta_{ab}\delta_{cd}\right). \tag{8.1}$$

With $D = \frac{1}{2}\Delta$ and E the polynomial Euler operator, the completeness identity for this basis gives the exact relations

$$DP_k = \frac{k}{2}\sum_{l=0}^{k-2} P_l P_{k-2-l} - \frac{k(k-1)}{2N} P_{k-2}, \tag{8.2}$$

$$D(fg) = (Df)g + f(Dg) + \nabla f \cdot \nabla g, \tag{8.3}$$

$$\nabla P_k \cdot \nabla P_l = k\,l\left(P_{k+l-2} - \frac{1}{N} P_{k-1} P_{l-1}\right). \tag{8.4}$$

These are exact at every integer rank, including ranks at which some traces become dependent — which is the property that makes the construction rank-uniform rather than rank-by-rank. For homogeneous F of positive degree d, Gaussian integration by parts gives $\mathbb{E}_\mu[F] = \mathbb{E}_\mu[DF]/d$, and with $\mathbb{E}_\mu[1] = 1$ and vanishing odd moments this is a finite exact moment recursion.

8.2 Shell projectors without a Gram inverse

Proposition 8.1 (Terminating Wick conjugation)**.** *Put $L = E - D$. Since $[E, D] = -2D$, the terminating exponential $W = \exp(-D/2)$ satisfies*

$$L\,W = W\,E. \tag{8.5}$$

Hence, with H_d the extraction of the homogeneous degree-d part,

$$\Pi_d = \exp(-D/2)\,H_d\,\exp(D/2) \tag{8.6}$$

is the exact oscillator-shell projector, and on each finite polynomial space $R_s = \sum_{d\neq s} \Pi_d/(s-d)$ is the exact shell resolvent.

The content of Proposition 8.1 is that L and E are conjugate by a *terminating* exponential, so the shell decomposition of the interacting operator is the shell decomposition of the Euler operator, transported. No Gram matrix is formed and none is inverted, which is precisely the step that makes rank-uniformity delicate elsewhere in this program.

8.3 The coefficients

Theorem 8.2 (All-rank local coefficients)**.** *The terminating conjugation gives exact shell projectors and a unique all-order simple-branch formal recursion in $\mathbb{Q}[N, N^{-1}]$, with state degree at most $s+4r$. The two formal local gaps have exact all-rank coefficients through β^{-2}, including c_3^- and $c_4^\pm$. The five corrections $c_0, \ldots, c_4$ are strictly negative at every allowed rank, by positive shifted numerator polynomials.*

Graph id: `RESULT:LOCAL_CLASS_WICK_RECURRENCE,RESULT:LOCAL_CLASS_WICK_COEFFICIENTS,RESULT:LOCAL_CLASS_WICK_SIGNS`

The construction applies to the vacuum, the first charge-even shell for $N \geq 2$, and the first charge-odd shell for $N \geq 3$. It recovers the previously archived c_0, c_1, c_2 formulas in both sectors and the archived all-rank c_3^+, and additionally derives c_3^- and both all-rank c_4 formulas. The certificate carries full polynomial eigen-equation residuals rather than only agreement with the archived constants — a distinction worth stating, because agreement with a previous value is a weaker check than satisfying the equation.

The recovery is by an independent route: the archive engine uses full-GUE cut-and-join and then subtracts the trace, while this one applies D directly in traceless coordinates.

8.4 A companion algebraic identity

Theorem 8.3 (Shifted-power carrier symbols)**.** *For the cubic Laurent operators with $S = L_\downarrow - 4I$,*

$$\sigma(RS^mR) = (q-4)^m(qe_2 + 3e_3) - 4\,\Pi_m(q)\,e_2^2 \tag{8.7}$$

for every natural m, with Π defined polynomially also at $q = 0$.

Graph id: `RESULT:RSM_R_CARRIER_SYMBOL_MASTER_THEOREM`

This closes the word table of §7 at all powers, and is formalized in Lean as an all-power operator statement.

Scope 8.4. Theorem 8.2 concerns the *formal* local class oscillator only. No error bound for the compact-group eigenvalues is claimed, and local spectral coefficients do not by themselves identify the interacting Wilson spectrum, an interacting-volume estimate, or a continuum source measure. Theorem 8.3 is an algebra identity and carries no sixth-order dynamical population claim.

9 External comparison

Published work enters this program as evidence rather than as authority. A paper is unchecked until something checks it, exactly like an internal document; the reason an external agreement counts for more is not that it is published but that it was produced without knowledge of this program, so agreement is not bookkeeping.

The map holds 108 indexed papers and 152 recorded relations to named claims here. 54 of those edges are verified — the paper was obtained, read, and compared against the claim — across 34 distinct papers. The remainder are recorded targets for papers not yet obtained, which is itself useful: a paper heavily cited within this web and still unobtained is automatically the next acquisition target.

An entry earns its place by naming a specific claim and saying what relationship it has to it. `workhouse lit -for G19` answers the query in the other direction, and validation rejects an entry whose target does not resolve.

9.1 A 1989 table, and rationals computed cold

The sharpest external check available to this program is a Hamiltonian strong-coupling series for SU(3) glueball masses published in 1989 [51], whose Table 1 is pinned here by digest. Under the normalization bridge $m_n = 2^{n-1}a_n$, its coefficients are compared against the Γ-point coefficients this program derives as exact rationals.

In the C-odd (1^{+-}) channel — the sector of the carrier, whose series is (2.5) — the agreement runs through third order:

Order	Published	This program	Gap (bound)
$n = 0$	$\frac{16}{3}/2$	$\frac{8}{3} = E_{\text{flat}}(0)$	exact
$n = 1$	$a_1 = +1$	the u coefficient $+1$	exact
$n = 2$	$2a_2 = 0.0359477124184$	$\frac{11}{306}$	9.94×10^{-14} (2×10^{-13})
$n = 3$	$4a_3 = -0.437135556836$	$-\frac{109151}{249696}$	1.11×10^{-12} (3×10^{-12})

In the C-even (0^{++}) channel, $2a_2 = -0.21274509804$ against $-\frac{217}{1020}$ with gap 7.84×10^{-13}, and $4a_3 = -0.1039004229144$ against $-\frac{54049}{520200}$ with gap 1.36×10^{-13}. The sign structure matches as well: $a_1 = -1$ in the scalar channel, where the carrier's is $+1$.

Each bound above is the printed half-ulp of the published table scaled with margin, so these are agreements to the precision at which the 1989 table was printed — not approximate corroborations. The rationals 11/306 and 109151/249696 are the third-order coefficients of E_{flat} in (2.5), computed here from Haar integration and the cellular complex with no knowledge of the table.

At fourth order, $8a_4 = -0.7751458630184$ against $m_\Gamma^{(4)} = -0.7751458630189173$, gap 5.17×10^{-13} against a bound of 5.3×10^{-13}.

Scope 9.1. These are [T2 numeric] checks: float agreement within a stated tolerance, to the precision of a printed table. They corroborate the coefficients; they do not prove them, and the exact statements of §3–§5 do not rest on them. Note also that the pinned revision of the source compares only $2a_2$ and $4a_3$ in its own text — the fourth-order agreement is registered here and not leaned on there.

9.2 The two contradicting edges

Two edges in the map are recorded as *contradicts*, both from a 1976 strong-coupling series, and both against this program: one against the third-order C-odd coefficient, one against the Γ-point fourth-order value. They originate in a disagreement the 1989 paper itself reports at orders x^3 and x^4 with the series it supersedes.

Both are recorded as not yet obtained. Until that primary source is read, this program has two unresolved external contradictions against coefficients it otherwise regards as settled, and says so rather than resolving them by the age of the sources. Obtaining that paper is listed with the acquisition targets below.

9.3 How far a novelty claim here can be trusted

Not as far as the size of the index suggests, and the limitation is worth stating precisely rather than as a disclaimer.

Only 54 of 152 recorded edges are verified. The unverified remainder is not a random sample: it includes several of the *nearest* neighbours to this program's own results. The main Nuclear Physics B paper of the 1981 strong-coupling glueball series is recorded as not obtained — only its errata were — and the 1974 and 1979 large-N papers that bear most directly on the rank grading of §5 are likewise unobtained, with the ledger noting for one of them that its definitions were used from memory and only their sign consequences checked.

Consequently: where this document says a result is novel, that means *no verified entry in this index contains it*, which is a weaker statement than priority. Two of the strongest results are affected. The rank-grading cancellation of §5 sits closest to large-N counting arguments in papers this program has not read; and the fixed-spacing transfer construction of §**??** sits alongside the classical positivity and self-adjoint transfer-matrix results of Osterwalder and Seiler, which the index does register.

Acquiring the unobtained near neighbours is cheap relative to everything else in §**??**, and it is the one item on that list that could *subtract* from the program rather than add to it. That is a reason to do it, not a reason to defer it.

Table 1: Verified external-evidence edges. Each row is a published result that was obtained, read, and checked against a named claim of this program. Relation is the ledger's own classification; *contradicts* and *confusable* are listed first because they are the rows that cost the program something.

Paper	Ref.	Year	Bears on	Relation
HSB_2000	[53]	2000	HAMER_A4_NUM	confusable with
HAMER_1989	[51]	1989	BAND_EVEN_BOTTOM	corroborates
HAMER_1989	[51]	1989	C1	corroborates
LLL_2006	[69]	2006	C1	corroborates
HAMER_1989	[51]	1989	D_3	corroborates
SCHIERHOLZ_1988	[91]	1988	G18	corroborates
HAMER_1989	[51]	1989	M3_EVEN_K0	corroborates
KPS_1981	[64]	1981	R14	corroborates
CBB_2026	[21]	2026	R2	corroborates
KPS_1981	[64]	1981	G7	supplies a value
HAMER_1989	[51]	1989	HAMER_A4_NUM	supplies a value
MUNSTER_1985_TM	[79]	1985	C2	supplies a comparison
CAO_2023	[16]	2023	C7	supplies a comparison
LEMOINE_2026	[68]	2026	C7	supplies a comparison
HAZRA_2026	[54]	2026	G14	supplies a comparison
AT_2020_SU3	[2]	2020	G18	supplies a comparison
BESIII_2026	[22]	2026	G18	supplies a comparison
CHEN_2006	[18]	2006	G18	supplies a comparison
LLL_2006	[69]	2006	G18	supplies a comparison
MORNINGSTAR_2025	[76]	2025	G18	supplies a comparison
MP_1999	[77]	1999	G18	supplies a comparison
AT_2021_SUN	[3]	2021	G19	supplies a comparison
BB_1983	[10]	1983	G19	supplies a comparison
FLPS_2026	[27]	2026	G19	supplies a comparison
URZUA_2019	[102]	2019	G19	supplies a comparison
BORGA_2024	[11]	2024	G5	supplies a comparison
CAO_2023	[16]	2023	G5	supplies a comparison
OBZ_1985	[82]	1985	G5	supplies a comparison
AT_2021_SUN	[3]	2021	G6	supplies a comparison
CM_2003	[17]	2003	G6	supplies a comparison
BALAJI_2026	[8]	2026	R2	supplies a comparison
CKS_2021	[19]	2021	U1	supplies a comparison
DRS_2021	[25]	2021	U1	supplies a comparison
KRS_2023	[62]	2023	U1	supplies a comparison
SZH_1997	[92]	1997	U1	supplies a comparison
CS_2006	[23]	2006	C7	supplies a method
JW_2006	[61]	2006	G17	supplies a method
UELTSCHI_2004_CLUSTER	[101]	2004	G17	supplies a method
YAROTSKY_2004_GS	[107]	2004	G17	supplies a method
BB_1983	[10]	1983	G18	supplies a method

Paper	Ref.	Year	Bears on	Relation
JW_2006	[61]	2006	G18	supplies a method
NSY_2019	[80]	2019	G18	supplies a method
YAROTSKY_2004_QP	[108]	2004	G18	supplies a method
JW_2006	[61]	2006	G19	supplies a method
KLEIN_ROSENBERGER_2020_WKB	[63]	2020	G19	supplies a method
SCHIERHOLZ_1988	[91]	1988	G19	supplies a method
UELTSCHI_2004_CLUSTER	[101]	2004	G19	supplies a method
YAROTSKY_2004_GS	[107]	2004	G19	supplies a method
YAROTSKY_2004_QP	[108]	2004	G19	supplies a method
JW_2006	[61]	2006	G20	supplies a method
JW_2006	[61]	2006	G22	supplies a method
JW_2006	[61]	2006	G23	supplies a method
MUNSTER_1985_TM	[79]	1985	G3	supplies a method
KRS_2023	[62]	2023	R2	supplies a method

A Reproducibility

Every claim in this document that carries a [T0 Lean], [T1 exact] or [T2 numeric] badge can be re-established from a clean checkout in seconds to minutes. This appendix says how, and — more usefully — says what each kind of re-establishment does and does not prove.

A.1 From a clean checkout

```
git clone https://github.com/ats314/WORKHOUSE
cd WORKHOUSE
uv sync --all-extras --frozen
uv run --no-sync workhouse --help
```

Then:

```
make verify         # every registered check: T1 and T2, a few seconds
make lean           # T0: proof-check the Lean core (requires elan)
make status         # the contradiction and gap registers
workhouse verify --tier 1             # only the exact re-derivations
workhouse verify -v                   # every tolerance, printed
workhouse verify --only 'h_4^side'    # one claim, with its numbers
```

A single claim is the useful unit. `workhouse verify --only` takes a claim name and prints the numbers it establishes, which is what makes a disagreement arguable rather than merely reported. A certification a reader cannot cheaply reproduce is a claim of authority, not evidence, and the design of this repository takes that seriously enough to attach the reproducing command to every row of `CERTIFIED.md`.

A.2 Querying a claim

```
workhouse why C2                   # everything recorded about one id
```

```
workhouse why t_N                  # a constant: value, formalizations, sources
workhouse branches C2              # both sides of a dispute, side by side
workhouse derive C2 G3 --out f.md  # evidence chain, registered edges only
workhouse search '5/612'           # search by exact VALUE, not by name
workhouse lit --for G19            # published work bearing on a claim
workhouse brief G19 --json         # target snapshot with freshness
```

`search` matches by value rather than spelling, which matters because the join keys of this corpus are exact rationals and no natural-language query retrieves 109151/249696. It also carries two warnings a text search cannot: names the register forbids because they regenerate a dissolved contradiction, and names coined in this program that its own source corpus never uses.

A.3 What each tier actually establishes

[T0 Lean] — Lean 4, no sorry.

The statement is checked by the Lean kernel from explicit definitions, with axioms restricted to `propext`, `Classical.choice` and `Quot.sound`. The axiom set is exported and validated rather than asserted: `scripts/export_lean_dependencies.py` builds strictly and records elaborated constants and transitive axioms into `ledger/lean_dependencies.json`, and the loader checks these against declaration records and source fingerprints. Identifiers mentioned in comments are not proof dependencies.

What a T0 badge does *not* establish is importance. Several registered theorems formalize arithmetic evaluations of named rationals. The registry records for each theorem whether it proves an entire named check (`promotes`) or one ingredient of a larger statement (`lean_support`); supporting a statement never silently promotes the whole statement's tier.

[T1 exact] — exact re-derivation.

The statement is recomputed symbolically in exact arithmetic from stated definitions. Corpus rationals are `sympy.Rational`; values the corpus records only as floats carry a `_NUM` suffix and are Python floats. A float that reads as exact is the most dangerous defect available in this repository, and the boundary is guarded by a dedicated test.

[T2 numeric] — numerical agreement.

Float agreement within a tolerance the check itself prints. A T2 pass is that and nothing more. Tolerances are never widened to make a disagreement disappear; a disagreement is registered as an explicit finding that asserts the discrepancy.

[T3 analytic] — no whole-statement machine certification.

The default. An analytic result may be fully proved and still be T3: the badge reports machine coverage, not mathematical status. Status, evidence and tier are three independent fields in `ledger/results.yaml` for exactly this reason.

A.4 The one inference this apparatus does not license

A finite check certifies the statement it checks. It does not certify a general theorem of which that statement is an instance, and the converse also holds: a complete analytic proof is not weakened by absent Lean coverage.

This cuts both ways in practice, and both directions have caused errors here. A suite of exact checks passing at ranks $N = 3, 4, 5, 6$ is not a statement about all N — which is why the symbolic-N engine of §6 exists, and why its degree bound was retracted when one projection turned out to be uncounted. Conversely, a separate Lean library in this program's archive comprises seventy-one modules with no `sorry` and was found on audit to prove mostly tautologies; a theorem count is not a measure of coverage, and no count in this document should be read as one.

A.5 Regenerating this document

```
python paper/master/generate_apparatus.py
tectonic paper/master/workhouse_master.tex        # the monograph
tectonic paper/master/workhouse_band_paper.tex    # Part I, standalone
```

The generator reads `literature/index.yaml` and `ledger/\{gaps,results,theorems,contradictions\}.yaml` and writes the bibliography, the external-evidence table, the route register, the results table and the count macros. Every number the prose quotes is one of those macros. If a count here disagrees with the repository, the generator has not been re-run; the repository is right.

B Verification index

Table 2 lists every registered analytic result with its recorded status and evidence level. The two columns are independent fields and are meant to be read against each other: `proven` with `record-backed` evidence means the argument exists and the machine artefact does not.

Of 143 registered results, 140 are recorded `proven`, 1 `conditional`, 1 `disputed` and 1 `falsified`. The Lean tree carries 439 registered theorems with 0 occurrences of `sorry`.

A caution on both counts. 140 out of 143 is a high proportion because a claim is not registered here until its argument is written down; the register measures what has been recorded, not what fraction of the problem is solved. Likewise 439 theorems is a count of declarations, not of mathematical content — see §A, which explains why a theorem count is not a coverage measure and cites a case in this program's own archive where seventy-one `sorry`-free modules turned out to prove tautologies.

Table 2: Analytic results by recorded status and evidence level. *Status* is what the argument establishes; *evidence* is what artefact backs it. The two are deliberately independent: a result can be `proven` in status and `record-backed` in evidence, meaning the proof exists and the machine artefact does not.

Result	Status	Evidence
ANISOTROPY_THREE_COORDINATE_SHARP_BOUND	proven	analytic
ARCHIVE_CENTER_DEFECT_OBSTRUCTIONS	proven	analytic
ARCHIVE_CONCENTRATION_LOCALIZATION	proven	analytic
ARCHIVE_FINITE_RANGE_INVERSE	proven	analytic
ARCHIVE_GLOBAL_PAIRING_OBSTRUCTION	proven	analytic
ARCHIVE_HORIZONTAL_RESTRICTION_BOUNDARY	proven	analytic
ARCHIVE_LYAPUNOV_PATCHING	proven	analytic
ARCHIVE_REFLECTION_PERMANENCE	proven	analytic
ARCHIVE_SIGNED_SU2_DIFFUSION	proven	analytic

Result	Status	Evidence
ARCHIVE_SMOOTH_PROFILES	proven	analytic
ARCHIVE_SOURCE_TILT_AND_ROOTED_BOUNDS	proven	analytic
ASSEMBLED_Q4_COERCIVITY	proven	analytic
BALABAN_MULTISCALE_CAUCHY_SUMMABILITY	proven	analytic
COMMON_SPACE_CONTRACTIVE_DRIFT	proven	analytic
COMPACT_ROTOR_FAST_GAP	proven	analytic
CONDITIONAL_GRADIENT_REPAIR	proven	analytic
CONDITIONAL_QUANTUM_PATH_COVARIANCE	proven	analytic
CREATOR_PARENT_GAP	proven	analytic
CREATOR_VELOCITY_INVERSION	proven	analytic
DIMENSION_FIVE_OPERATOR_IRRELEVANCE	proven	analytic
DYNAMIC_CONDITIONAL_CUBIC_ENERGY	proven	analytic
EARLIER_B6_RETAINED_KRYLOV_CAMPAIGN	conditional	numerical
EARLIER_CONDITIONAL_SPECTRAL_FLOOR	proven	analytic
EARLIER_CORNER_BLOCKING_REFLECTION_DEFECT	proven	analytic
EARLIER_CUBIC_QUOTIENT_REGULARITY	proven	analytic
EARLIER_D4_DISJOINT_STAPLE_COORDINATES	proven	analytic
EARLIER_DECORATED_HISTORY_MERGING	proven	analytic
EARLIER_DETERMINANT_PARITY_REPAIR	proven	analytic
EARLIER_EQUAL_RAY_IDENTIFICATION	proven	analytic
EARLIER_FINITE_ORDER_SUPPORT	proven	analytic
EARLIER_GRAM_NULL_OPERATOR_QUOTIENT	proven	analytic
EARLIER_ISOLATED_POSITIVE_MATRIX_ATOM	proven	analytic
EARLIER_JOINT_RANK_VOLUME_SCALING	proven	analytic
EARLIER_LOCALIZED_FRAME_OBSTRUCTION	proven	analytic
EARLIER_MOVING_ADJOINT_FORCE_DERIVATIVE	proven	analytic
EARLIER_NESTED_QUOTIENT_REDUCTION	proven	analytic
EARLIER_POSITIVE_SECTOR_PHASE_ISOLATION	proven	analytic
EARLIER_REFLECTION_ADAPTED_BLOCKING	proven	analytic
EARLIER_SHELL_SYMMETRY_RITZ_CONTROL	proven	analytic
EARLIER_SPHERE_EQUAL_COUPLING_COVARIANCE	proven	analytic
EARLIER_SU3_POSITIVITY_BOUNDARIES	proven	analytic
EARLIER_VSU_BOUNDED_DOMAIN_WELLPOSEDNESS	proven	analytic
EXPONENTIAL_SOURCE_TOTALITY	proven	analytic
FESHBACH_RESOLVENT_COMPARISON	proven	analytic
FORM_SCHUR_SCALE_COMPARISON	proven	analytic
FULL_GRAPH_SOURCE_CONGRUENCE	proven	analytic
G17_CORRECTED_KP_POLYMER_STABILITY	proven	analytic
G9_COMBINED_MIXING_SUPPORT	proven	analytic
G9_HERMITIAN_FOLDED_SUPPORT	proven	analytic
G9_LOCAL_SIXTH_NONCANCELLATION	proven	analytic
G9_ONE_FACE_SIXTH	proven	analytic
G9_SIXTH_ORDER_RUR_ACTIVATION	disputed	record-backed
GAUSSIAN_FULL_FAST_GREEN	proven	analytic
GAUSSIAN_OS_OBSERVABILITY	proven	analytic
GAUSSIAN_PATH_ENDPOINT_BASELINE	proven	analytic

Result	Status	Evidence
GAUSSIAN_QUANTUM_FAST_SOURCES	proven	analytic
GAUSSIAN_SCHUR_MEMORY	proven	analytic
GROUND_MARGINAL_SCHUR_SCORE	proven	analytic
HODGE_FESHBACH_SPLITTING	proven	analytic
HODGE_INDUCED_EIGHTH_ORDER_MOMENT	proven	analytic
HODGE_LEAKAGE_ZERO_EXTENSION	proven	analytic
HODGE_RANK_ONE_SECULAR_CUBIC	proven	analytic
INTRINSIC_SCORE_CURRENT	proven	analytic
LITERAL_ENDPOINT_SPECTRAL_COMPARISON	proven	analytic
LOCAL_CLASS_WICK_COEFFICIENTS	proven	analytic
LOCAL_CLASS_WICK_RECURRENCE	proven	analytic
LOCAL_CLASS_WICK_SIGNS	proven	analytic
LOCAL_GRADIENT_EXCITATION_SUPPORT	proven	analytic
MESOSCOPIC_NORMALIZED_SOURCE	proven	analytic
MOVING_TIME_MULTICHANNEL_RATE	proven	analytic
MOVING_TIME_MULTICHANNEL_SHARPNESS	proven	analytic
MOVING_TIME_SPECTRAL_GAP	proven	analytic
ODD_ORDER_CENTRE_PARITY	proven	analytic
ORDERED_CONTOUR_ACTIVITIES	proven	analytic
OS_HISTORY_BLOCK_INTERTWINER	proven	analytic
OS_KERNEL_MOVING_TIME_GAP	proven	analytic
PREDICTABLE_SOURCE_DRIFT	proven	analytic
PROJECTIVE_SOURCE_MARTINGALE	proven	analytic
REVERSE_MARTINGALE_BLOCKING	proven	analytic
RICCATI_RESIDUAL_CERTIFICATE	proven	analytic
ROUGH_GAUGE_COERCIVITY_LOWER_BOUND	proven	analytic
RSM_R_CARRIER_SYMBOL_MASTER_THEOREM	proven	analytic
SELECTED_INVERSE_WITHOUT_DIAGONAL	proven	analytic
SHARP_CUTOFF_SOURCE_GAP_BUDGET	proven	analytic
SOURCE_COMPLEMENT_COMPLETE_WINDOW	proven	analytic
SOURCE_CONGRUENCE_METRIC_TRANSPORT	proven	analytic
SOURCE_CURRENT_BOUNDED_W6	proven	analytic
SOURCE_CURRENT_VARIATIONAL_ENERGY	proven	analytic
SQUARE_FIRST_CURRENT_BUDGET	proven	analytic
SQUARE_FIRST_RESIDUAL_CURRENT	proven	analytic
SQUARE_SCALAR_SOURCE_ODD_SCHUR_VANISHING	proven	analytic
TETRAHEDRAL_FLUX_PROJECTION_BOUNDARY	proven	analytic
TETRAHEDRAL_S4_COMMUTANT_RESOLUTION	proven	analytic
TIER_COLLAPSE_ACTUAL_H4_SUPPORT	proven	analytic
TIER_COLLAPSE_IS_R_DEGREE	falsified	analytic
TRUE_GROUND_CENTER_SCORE_OBSTRUCTION	proven	analytic
UNIFORM_MARKED_CLUSTERING	proven	analytic
UNIVERSAL_CELLULAR_HODGE_SPECTRUM	proven	analytic
W6_ACTUAL_CENTERED_COMPLEMENT_SUPPRESSION	proven	analytic
W6_ANGULAR_REFERENCE_SCORE_BUDGET	proven	analytic
W6_R10_CONDITIONAL_DERIVATIVE	proven	analytic

Result	Status	Evidence
W6_R10_H4_FROM_H0	proven	analytic
W6_R10_PROJECTION_JET_RECURRENCE	proven	analytic
W6_UNIFORM_AGMON_COMPLEMENT_GAP	proven	analytic
WILSON_ACTIVITY_EXTRACTION	proven	analytic
WILSON_ACTUAL_BLOCK_FAST_COMPLEMENT	proven	analytic
WILSON_BLOCK_SCORE_OBSTRUCTION	proven	analytic
WILSON_CARDINALITY_CHART	proven	analytic
WILSON_COEFFICIENT_LOCALITY	proven	analytic
WILSON_COMMON_GAUSS_LITERAL_FAST_COMPLEMENT	proven	analytic
WILSON_COVARIANT_BLOCK_SOURCES	proven	analytic
WILSON_CREATOR_LIMIT	proven	analytic
WILSON_CUBIC_GROUND_TRANSFER	proven	analytic
WILSON_ENDPOINT_COMPLETE_CLUSTER	proven	analytic
WILSON_ENDPOINT_EQUATION	proven	analytic
WILSON_FINITE_CELL_PHYSICAL_GAP	proven	analytic
WILSON_FIXED_ORDER_CHART	proven	analytic
WILSON_FLAT_BACKGROUND_FAST_SOURCES	proven	analytic
WILSON_GLOBAL_VERTICAL_BARRIER	proven	analytic
WILSON_GROUND_BUNDLE_RELATIVE_FORM	proven	analytic
WILSON_HARMONIC_BOUNDARY	proven	analytic
WILSON_HARMONIC_CUBIC_OBSTRUCTION	proven	analytic
WILSON_INFINITE_PHYSICAL_BAND	proven	analytic
WILSON_KINETIC_WINDOW	proven	analytic
WILSON_LITERAL_COARSE_FORM	proven	analytic
WILSON_LITERAL_FAST_COMPLEMENT	proven	analytic
WILSON_LOCALIZED_TRUE_GROUND_SCORE	proven	analytic
WILSON_PARENT_GNS	proven	analytic
WILSON_PHYSICAL_SOURCE_RANK_REPAIR	proven	analytic
WILSON_ROOTED_CONTRACTION	proven	analytic
WILSON_SAME_WEIGHT_OBSTRUCTION	proven	analytic
WILSON_SECOND_ORDER_CHART	proven	analytic
WILSON_STRIP_B0_FIRST_TERM	proven	analytic
WILSON_SYMMETRIC_CREATORS	proven	analytic
WILSON_THREE_DIMENSIONAL_HARMONIC_BOUNDARY	proven	analytic
WILSON_TRANSFER_RESOLVENT	proven	analytic
WILSON_TWO_SQUARE_PHYSICAL_SHELLS	proven	analytic
WILSON_TWO_STRIP_PHYSICAL_SPLITTING	proven	analytic
WILSON_UNIFORM_FINITE_SHELL	proven	analytic
WILSON_VACUUM_COMPRESSION	proven	analytic
WILSON_VACUUM_SPECTRAL_FLOW	proven	analytic
WILSON_VERTICAL_FAST_ENERGY	proven	analytic
WILSON_WEIGHTED_ACTIVITIES	proven	analytic

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